

## TONOPAH, NV, USA

WMO: 724855

Lat: 38.051N

Lon: 117.090W

Elev: 1644

StdP: 83.07

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 23153

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 12            | -11.6      | -9.2 | -22.5                         | 0.6 | 0.2  | -20.1 | 0.8 | 1.8  | 14.2                  | 2.5  | 12.5 | 1.8  | 4.0                   | 0    | 0.619 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 17.6                   | 35.3            | 15.1 | 34.1 | 14.6 | 32.9 | 14.1 | 17.3                | 29.8 | 16.4 | 28.7 | 15.7 | 28.3 | 5.0                  | 200  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 13.9                            | 12.1 | 19.9 | 12.3 | 10.9 | 20.5 | 10.4 | 9.6 | 21.4 | 54.6          | 29.4 | 51.8 | 28.1 | 49.4 | 28.3 | 24.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 12.5          | 11.0 | 9.4 | DB                         | -16.0 | 37.6               | 2.7 | 0.9                                                | -17.9 | 38.3       | -19.5 | 38.9       | -21.0 | 39.4       | -22.9 | 40.1 | (4) |
| (5)               |      |     | WB                         | -16.8 | 18.7               | 2.4 | 1.5                                                | -18.5 | 19.7       | -19.9 | 20.6       | -21.2 | 21.4       | -22.9 | 22.5 | (5) |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 11.6  | 1.2   | 2.7   | 6.6   | 9.8   | 14.8  | 20.7  | 24.3  | 23.1  | 18.6  | 11.7  | 5.2   | 0.3  | (6)  |
| (7)  |                                                                                  | DBStd       | 9.09  | 4.04  | 4.04  | 3.95  | 4.28  | 4.34  | 3.97  | 2.40  | 2.24  | 3.38  | 3.93  | 4.18  | 3.83 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 1140  | 272   | 204   | 117   | 56    | 11    | 1     | 0     | 0     | 1     | 28    | 150   | 301  | (8)  |
| (9)  |                                                                                  | HDD18.3     | 2932  | 530   | 437   | 363   | 257   | 123   | 23    | 0     | 1     | 37    | 207   | 394   | 559  | (9)  |
| (10) |                                                                                  | CDD10.0     | 1737  | 0     | 1     | 12    | 49    | 161   | 322   | 443   | 406   | 258   | 81    | 6     | 0    | (10) |
| (11) |                                                                                  | CDD18.3     | 486   | 0     | 0     | 0     | 0     | 14    | 94    | 184   | 148   | 43    | 1     | 0     | 0    | (11) |
| (12) |                                                                                  | CDH23.3     | 7764  | 0     | 0     | 2     | 33    | 363   | 1593  | 2721  | 2182  | 801   | 69    | 1     | 0    | (12) |
| (13) |                                                                                  | CDH26.7     | 3684  | 0     | 0     | 0     | 2     | 98    | 750   | 1472  | 1090  | 265   | 8     | 0     | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 4.3   | 3.9   | 4.4   | 4.7   | 5.3   | 4.9   | 4.6   | 4.1   | 4.0   | 4.2   | 4.0   | 3.9   | (14) |      |
| (15) | Precipitation                                                                    | PrecAvg     | 135   | 9     | 15    | 12    | 9     | 14    | 10    | 11    | 12    | 14    | 10    | 5     | (15) |      |
| (16) |                                                                                  | PrecMax     | 270   | 27    | 68    | 60    | 44    | 52    | 42    | 43    | 57    | 52    | 55    | 43    | 26   | (16) |
| (17) |                                                                                  | PrecMin     | 74    | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    | (17) |
| (18) |                                                                                  | PrecStd     | 49    | 8     | 17    | 15    | 10    | 15    | 13    | 12    | 13    | 16    | 13    | 11    | 6    | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 15.2  | 18.4  | 23.0  | 26.6  | 32.0  | 36.1  | 37.3  | 35.9  | 32.5  | 27.7  | 21.4  | 14.1 | (19) |
| (20) |                                                                                  |             | MCWB  | 5.5   | 6.6   | 8.5   | 10.4  | 13.7  | 14.7  | 15.9  | 15.9  | 13.5  | 11.7  | 8.5   | 5.3  | (20) |
| (21) |                                                                                  | 2%          | DB    | 12.9  | 15.7  | 20.2  | 24.0  | 29.2  | 34.2  | 36.0  | 34.5  | 30.9  | 25.2  | 18.9  | 11.8 | (21) |
| (22) |                                                                                  |             | MCWB  | 4.4   | 5.5   | 7.2   | 9.0   | 12.1  | 14.0  | 15.5  | 15.0  | 13.2  | 10.6  | 7.3   | 4.0  | (22) |
| (23) |                                                                                  | 5%          | DB    | 11.0  | 13.3  | 18.0  | 22.1  | 27.3  | 32.6  | 34.7  | 33.3  | 29.7  | 23.1  | 16.2  | 9.8  | (23) |
| (24) |                                                                                  |             | MCWB  | 3.7   | 4.4   | 6.3   | 8.2   | 11.2  | 13.6  | 14.9  | 14.4  | 12.7  | 9.3   | 6.1   | 2.9  | (24) |
| (25) |                                                                                  | 10%         | DB    | 8.7   | 10.6  | 15.9  | 19.8  | 25.2  | 31.2  | 33.4  | 32.2  | 28.1  | 21.1  | 13.5  | 7.6  | (25) |
| (26) |                                                                                  |             | MCWB  | 2.6   | 3.3   | 5.4   | 7.3   | 10.2  | 12.9  | 14.5  | 13.9  | 12.3  | 8.5   | 4.8   | 2.0  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 7.5   | 7.6   | 9.1   | 10.9  | 14.9  | 17.7  | 19.3  | 18.6  | 16.4  | 13.0  | 9.6   | 6.8  | (27) |
| (28) |                                                                                  |             | MCDWB | 12.0  | 15.5  | 20.1  | 24.8  | 30.5  | 32.7  | 31.8  | 32.3  | 26.7  | 22.2  | 18.5  | 11.6 | (28) |
| (29) |                                                                                  | 2%          | WB    | 5.7   | 6.3   | 7.8   | 9.7   | 12.9  | 15.6  | 17.6  | 16.9  | 15.4  | 11.6  | 7.9   | 5.2  | (29) |
| (30) |                                                                                  |             | MCDWB | 10.0  | 14.2  | 18.4  | 22.7  | 26.7  | 32.0  | 29.9  | 29.5  | 26.2  | 22.2  | 17.0  | 9.3  | (30) |
| (31) |                                                                                  | 5%          | WB    | 4.5   | 5.1   | 6.8   | 8.7   | 11.7  | 14.4  | 16.7  | 16.0  | 14.4  | 10.4  | 6.7   | 3.8  | (31) |
| (32) |                                                                                  |             | MCDWB | 9.2   | 11.5  | 16.9  | 20.6  | 25.4  | 30.7  | 28.8  | 28.5  | 25.5  | 21.2  | 15.2  | 8.5  | (32) |
| (33) |                                                                                  | 10%         | WB    | 3.2   | 3.9   | 5.8   | 7.7   | 10.6  | 13.3  | 16.0  | 15.3  | 13.3  | 9.2   | 5.3   | 2.3  | (33) |
| (34) |                                                                                  |             | MCDWB | 8.2   | 9.4   | 14.9  | 18.9  | 23.6  | 29.0  | 28.2  | 27.9  | 24.9  | 19.3  | 12.5  | 6.8  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 12.8  | 12.9  | 15.0  | 15.2  | 15.8  | 17.3  | 17.6  | 17.7  | 16.9  | 15.9  | 14.5  | 13.0 | (35) |
| (36) |                                                                                  |             | MCDBR | 16.3  | 17.0  | 18.2  | 18.5  | 18.7  | 18.9  | 18.8  | 18.5  | 18.3  | 18.6  | 17.9  | 16.0 | (36) |
| (37) |                                                                                  |             | MCWBR | 9.5   | 9.3   | 9.1   | 8.8   | 8.6   | 8.5   | 8.0   | 8.0   | 8.3   | 9.4   | 9.8   | 9.9  | (37) |
| (38) |                                                                                  | 5% WB       | MCDWB | 13.4  | 15.0  | 17.1  | 17.4  | 17.8  | 18.5  | 16.3  | 16.8  | 16.4  | 16.9  | 16.6  | 14.2 | (38) |
| (39) |                                                                                  |             | MCWBR | 8.3   | 8.6   | 8.9   | 8.5   | 8.2   | 9.2   | 6.7   | 7.3   | 7.9   | 8.7   | 9.3   | 9.0  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.238 | 0.245 | 0.254 | 0.287 | 0.300 | 0.298 | 0.316 | 0.309 | 0.283 | 0.259 | 0.242 | 0.236 | (40) |      |
| (41) |                                                                                  | taud        | 2.651 | 2.636 | 2.611 | 2.479 | 2.474 | 2.498 | 2.493 | 2.492 | 2.565 | 2.640 | 2.658 | 2.662 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 981   | 1010  | 1022  | 994   | 978   | 976   | 955   | 958   | 974   | 977   | 968   | 961   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 66    | 77    | 87    | 106   | 109   | 107   | 107   | 104   | 91    | 76    | 65    | 61    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 2.73  | 3.69  | 5.19  | 6.56  | 7.42  | 8.42  | 7.72  | 7.17  | 6.10  | 4.47  | 3.17  | 2.39  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.26  | 0.37  | 0.30  | 0.34  | 0.57  | 0.52  | 0.41  | 0.41  | 0.24  | 0.29  | 0.14  | 0.20  | (45) |      |

## Historical Trends

|      |                        | Heating |        | Cooling |       |       | Degree-Days |         |         |         |     |      |
|------|------------------------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|-----|------|
|      |                        | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |     |      |
|      |                        | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     | (m) |      |
| (46) | Station Only           | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     | N/A | (46) |
| (47) | Regional (7 neighbors) | +0.50   | N/A    | -1.02   | N/A   | N/A   | N/A         | N/A     | N/A     | +130    | +64 | (47) |

Nomenclature: See separate page